

Multiple Choice (10 marks)

1. If the finite group G contains a subgroup of order five but no element of G other than the identity is its own inverse, then the order of G could be
 - (a) 8
 - (b) 20
 - (c) 30
 - (d) 35
2. If A is a subset of the real line $\mathbb{R}$ and A contains each rational number, which of the following must be true?
 - (a) If A is open, then $A = \mathbb{R}$.
 - (b) If A is closed, then $A = \mathbb{R}$.
 - (c) If A is uncountable, then $A = \mathbb{R}$.
 - (d) If A is uncountable, then A is open.
3. What is the volume of the solid in xyz -space bounded by the surfaces $y = x^2$, $y = 2 - x^2$, $z = 0$, and $z = y + 3$?
 - (a) $8/3$
 - (b) $16/3$
 - (c) $32/3$
 - (d) $104/105$
4. Let $x_1 = 1$ and $x_{n+1} = \sqrt{3 + 2x_n}$ for all positive integers n . If it is assumed that $\{x_n\}$ converges, then $\lim x_n =$
 - (a) 3
 - (b) e
 - (c) $\sqrt{5}$
 - (d) 0
5. Statement 1 | A ring homomorphism is one to one if and only if the kernel is $\{0\}$.
Statement 2 | $\mathbb{Q}$ is an ideal in $\mathbb{R}$.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Every field is also a ring because it satisfies all ring axioms including the properties of addition and multiplication.
7. True or False: Every group of order 42 contains a normal subgroup of order 7 due to Sylow's theorems.
8. True or False: In the ring of integers modulo 11, the product of two nonzero elements can be zero.
9. True or False: If f is a linear transformation and $f(1, 1) = 1$, then $f(3, 5)$ equals 9.
10. True or False: Every field is an integral domain, as it satisfies the properties of both a field and an integral domain.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There exist two distinct unit vectors $\mathbf{v}$ such that the angle between $\mathbf{v}$ and $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ is 45° , and the angle between $\mathbf{v}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ is 60° . Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be these vectors. Find $\|\mathbf{v}_1 - \mathbf{v}_2\|$.
12. Discuss the possible dimensions of the intersection $U \cap V$ when U and V are 3-dimensional subspaces of $\mathbb{R}^5$. Provide a detailed explanation of your reasoning and any relevant concepts.

End of Paper